

## TRACK B / TECHNICAL RESEARCH AND PROTOTYPING

# Find the worthwhile automation

Investigate a business problem and a technical claim, then test your recommendation. This shared case assesses research and prototyping skills applicable across technology sectors.

## The situation

Daybreak Repairs is a fictional appliance maintenance business with four technicians and one coordinator. Its owner says: "We lose eight hours a week chasing customers. Can AI fix this?"

## The decision

Choose a narrow problem around **collecting missing information or getting quote approval**. Should the business build something, buy a tool, change its process or leave the problem alone for now?

The scenario allows up to **two engineering weeks** and a recurring tool budget of **INR 1,500/month**. These are constraints, not evidence of real willingness to pay. Do not assume the owner's eight-hour estimate is true.

## Deliver three things

- 1. A decision note.** In 500-700 words, explain the problem, one consequential technical claim, evidence, alternatives, recommendation, counterargument and next experiment.
- 2. A working experiment.** Directly test your claim using supplied data and compare with a simple baseline. Include two meaningful checks and a valid no-action input; one check can cover both an edge case and a changed input.
- 3. Sources and handover.** Give three to five useful external sources with direct links, access dates and limitations. Include exact run commands, tool decisions, time spent and what remains unknown.

## Seven calendar days to submit. Four hours of work.

Choose one track. Include reading, choosing, research, coding, checks and handover in the four-hour limit. You may split the work across sessions. Any model or free tool is fine. No paid API, customer interview or deployment is required.

# Investigate the evidence

Start with the files in **track-b**. Everything in this case study is synthetic. Use the fixed snapshot: **7 September 2026, 09:00 IST**, not your computer's current clock.

| Evidence           | Use it to investigate                                                           |
|--------------------|---------------------------------------------------------------------------------|
| data/cases.csv     | Case status, service type, channel and quoted price                             |
| data/events.csv    | Activity deliveries and known coordinator effort; inspect duplicates and blanks |
| data/requests.csv  | Missing information and approvals, including stale or conflicting records       |
| USER_NOTES.md      | Six fictional perspectives that may disagree                                    |
| DATA_DICTIONARY.md | Field meanings, limits, joins and required prototype rules                      |

## Make a defensible decision

Show **two or three concrete calculations**. Explain duplicate handling, missing values and inconsistent states. Waiting time is not staff effort. A quoted price is not earned revenue or profit.

Compare **at least two alternatives**, including an existing tool and a process-only change. Check relevant capabilities, constraints and cost assumptions. Use **RESOURCE\_STARTERS.md** if helpful; no vendor is preferred.

Investigate **one consequential technical claim**: what must be true for your method or tool to work? Check original technical documentation or research. Identify the key constraint and separate demonstrated capability from assumptions.

Estimate **net value for your selected workflow**: assumed time saved minus review, reconciliation and operating time, with relevant tool costs. Label missing measurements and assumptions. Negative or uncertain value is acceptable. Give contrary evidence and a measurable continue/stop criterion.

## Build a narrow experiment

A comparison or check must **directly test the technical claim**. Separate local mock results from documentation-only claims. Compare a **simple baseline** on the same inputs and measure. Predict a changed input's effect and record the result. Include valid input with **no proposed actions** (or no work for a non-contact workflow) without crashing. Your method need not win.

### If your experiment proposes contact

Dry run only. Exclude closed, scheduled, opted-out and already-received items. Wait at least 48 elapsed hours since the last request. Missing or conflicting fields require review. Use stable IDs and explain actions; reruns must not accumulate duplicates. Read the complete rules in **DATA\_DICTIONARY.md**.

# Prove enough to decide

Your experiment should run without credentials, produce a useful output and respond meaningfully when an input changes. State what it proves and what it does not.

| Criterion                           | Weight | What we look for                                                                  |
|-------------------------------------|--------|-----------------------------------------------------------------------------------|
| Research and source quality         | 30%    | Checked business and technical evidence, alternatives and a questioned premise    |
| Analysis and reasoning              | 25%    | Correct calculations, explicit assumptions and a defensible decision              |
| Working experiment and verification | 25%    | Runnable test of the claim, baseline comparison, changed inputs and honest limits |
| Scope and prioritization            | 10%    | A feasible next step and a clear continue/stop criterion                          |
| Handover and tool judgment          | 10%    | Clear communication and ownership of tool output                                  |

## Optional starter

From **track-b**, run the correct Python 3.10+ loader below. It has no dependencies and is not a bug hunt. Extend it or use another language.

```
python starter.py
```

## A suggested time budget

**35 min** inspect data / **50 min** targeted research / **35 min** analyze

**85 min** build and check / **35 min** decision note and handover

## Your submission

Within **seven calendar days of the invitation**, reply with an accessible download link to **YourName\_TrackB.zip**, or a repository link. Include code, checks, output, **DECISION.md**, **SOURCES.md** and **HANDOVER.md**. The note is 500-700 words excluding tables/calculations; keep the handover under 500 words excluding commands.

Use three to five sources with URL, access date, supported claim and limitation. At least one must be original technical documentation or research. Credit outside code and describe two or three tool decisions and checks. Preserve source data; exclude credentials/caches. Check the inviter can access your link.

Work independently within the requirements; document assumptions and unresolved questions. Contact the inviter only for inaccessible/missing materials or an accessibility adjustment. Shortlisted applicants will explain and adapt their work in a separate **15-20 minute discussion**, with AI allowed. See **START\_HERE.md** and the track README.